

2(a)(i)	4 / four ;	1								
2(a)(ii)	COOH group bonded to the top carbon ; NH ₂ group bonded to the top carbon ; A with or without lines indicating bonds within functional groups R if atoms other than C and N are shown bonded to the carbon	2								
2(b)	<table><tr><th>description</th><th>level of protein structure</th></tr><tr><td>in some conditions, four melittin polypeptides can bind to each other</td><td>quaternary ;</td></tr><tr><td>a melittin polypeptide consists of a sequence of 26 amino acids</td><td>primary ;</td></tr><tr><td>alpha helices are formed at each end of a melittin polypeptide</td><td>secondary ;</td></tr></table>	description	level of protein structure	in some conditions, four melittin polypeptides can bind to each other	quaternary ;	a melittin polypeptide consists of a sequence of 26 amino acids	primary ;	alpha helices are formed at each end of a melittin polypeptide	secondary ;	3
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2(c)	produces a, pore / gap / channel / opening / AW, (in cell surface membrane) ; A forms a channel protein <i>ref. to phospholipids / phospholipid bilayer ; e.g. displaces / disrupts / separates, phospholipids</i>	2								

Question	Answer	Marks
2(d)	<i>any four from:</i> 1 binding / attachment / joining / AW, of cell fragment to, (phagocyte cell surface) receptors / (phagocyte) membrane ; <i>endocytosis / phagocytosis</i> 2 (cell surface) membrane, surrounds / AW, cell fragment or pseudopodia, surrounds / form round / AW, cell fragment or (phagocytic) cell, envelops / engulfs, cell fragment ; 3 membrane fusion / (phagocytic), vacuole pinches off / AW or phagocytic vacuole formed ; <i>allow vacuole if phagocytosis stated</i> A phagosome formed A vesicle for vacuole <i>break down cell fragments</i> 4 lysosome (containing enzymes), fuses / AW, with phagocytic vacuole ; A phagolysosome formed 5 fragment broken down by, hydrolytic / digestive, enzymes ; R lysosome digests 6 named enzyme and product or two named enzymes ; AVP ; <i>idea of opsonisation of / antibody binding to, cell fragment</i>	4